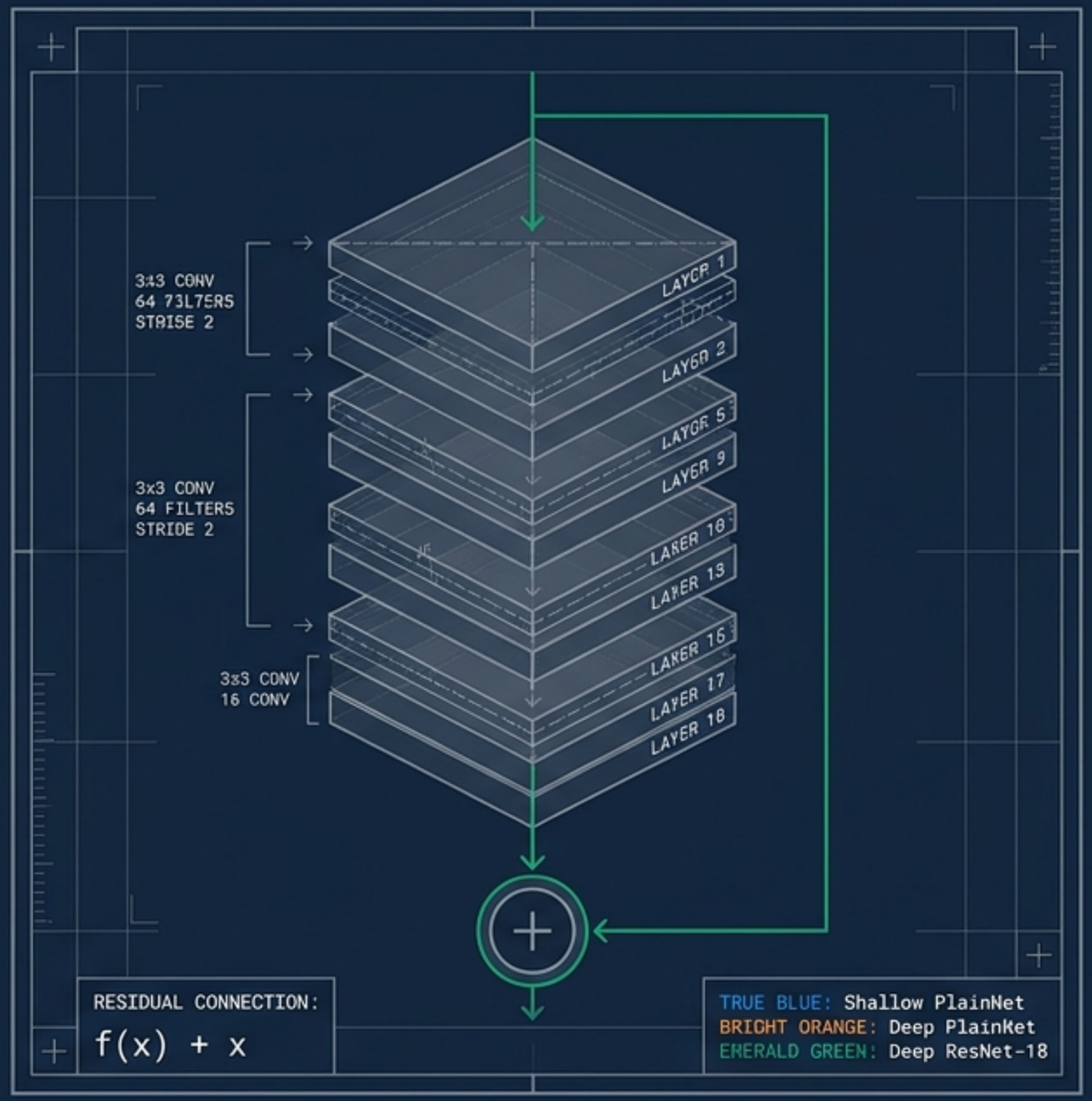
ResNet-18 & The Degradation Proof

An empirical investigation into network depth, optimization bottlenecks, bottlenecks, and residual learning on CIFAR-10.



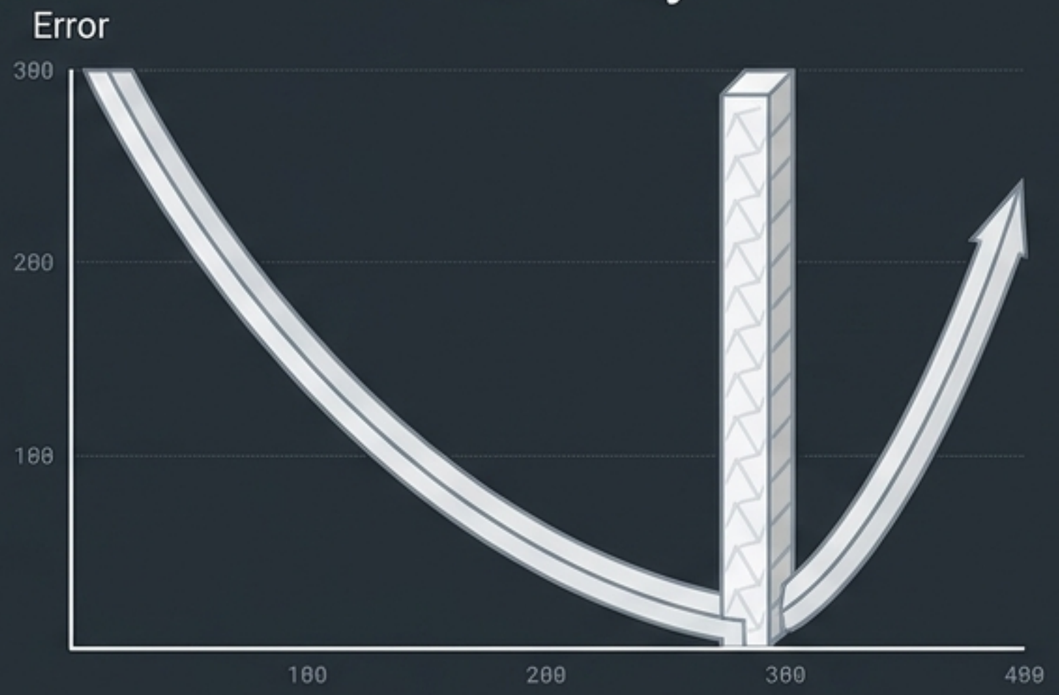
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The Assumption



Deeper Network = Better Abstraction & **Lower Error**

The Reality

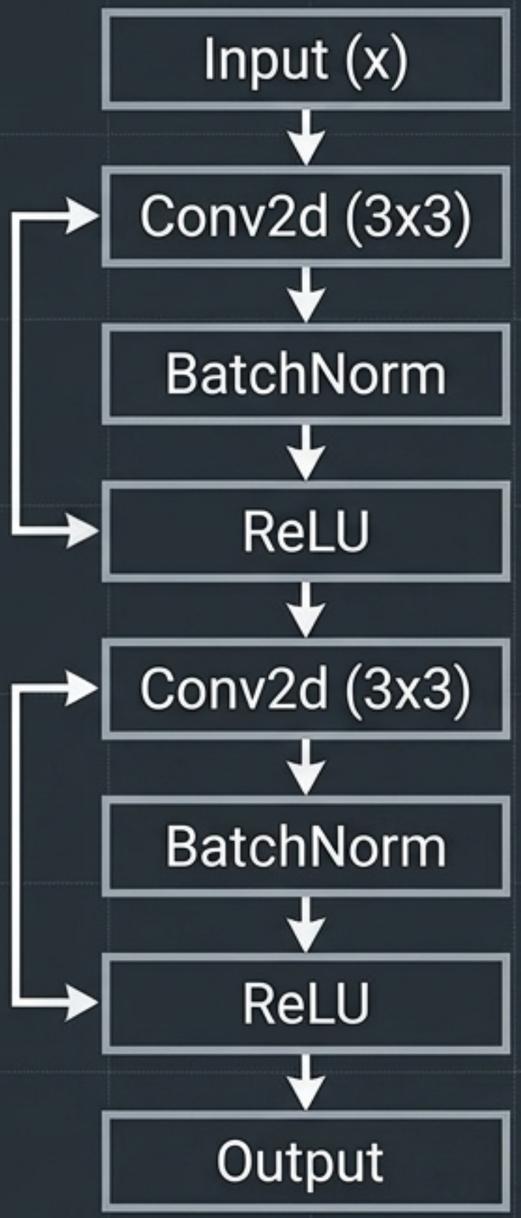


Deeper Network = **Higher Training Error**

Degradation is not overfitting.

It is an **optimization bottleneck**. When adding depth to a plain network, it eventually becomes harder to optimize, causing training error to degrade long before the model has a chance to overfit the data.

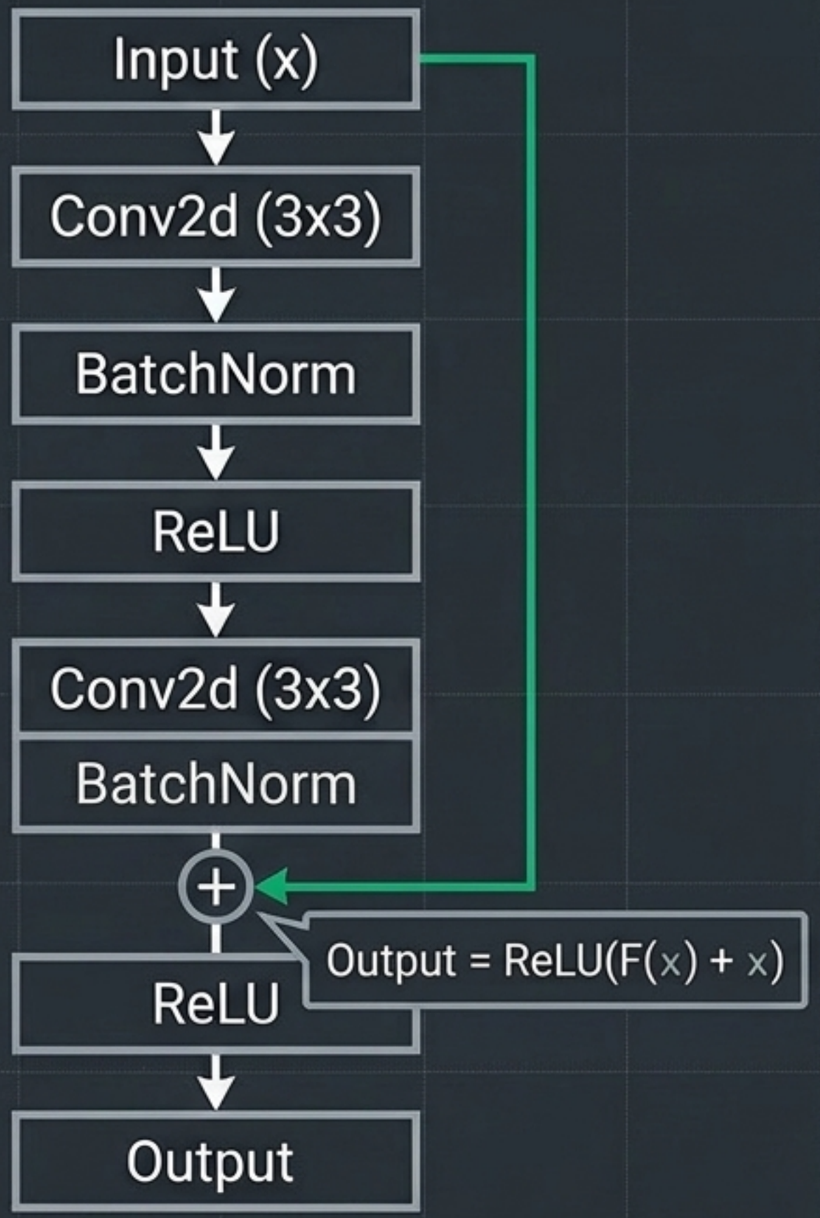
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The Plain CNN Block: The Straight Path

- Every extracted feature must pass strictly through every convolutional layer.
- As layers stack, gradients must navigate this unbroken chain during backpropagation.
- No bypass mechanisms exist; if an identity mapping is optimal, the non-linear layers must manually learn to simulate it.

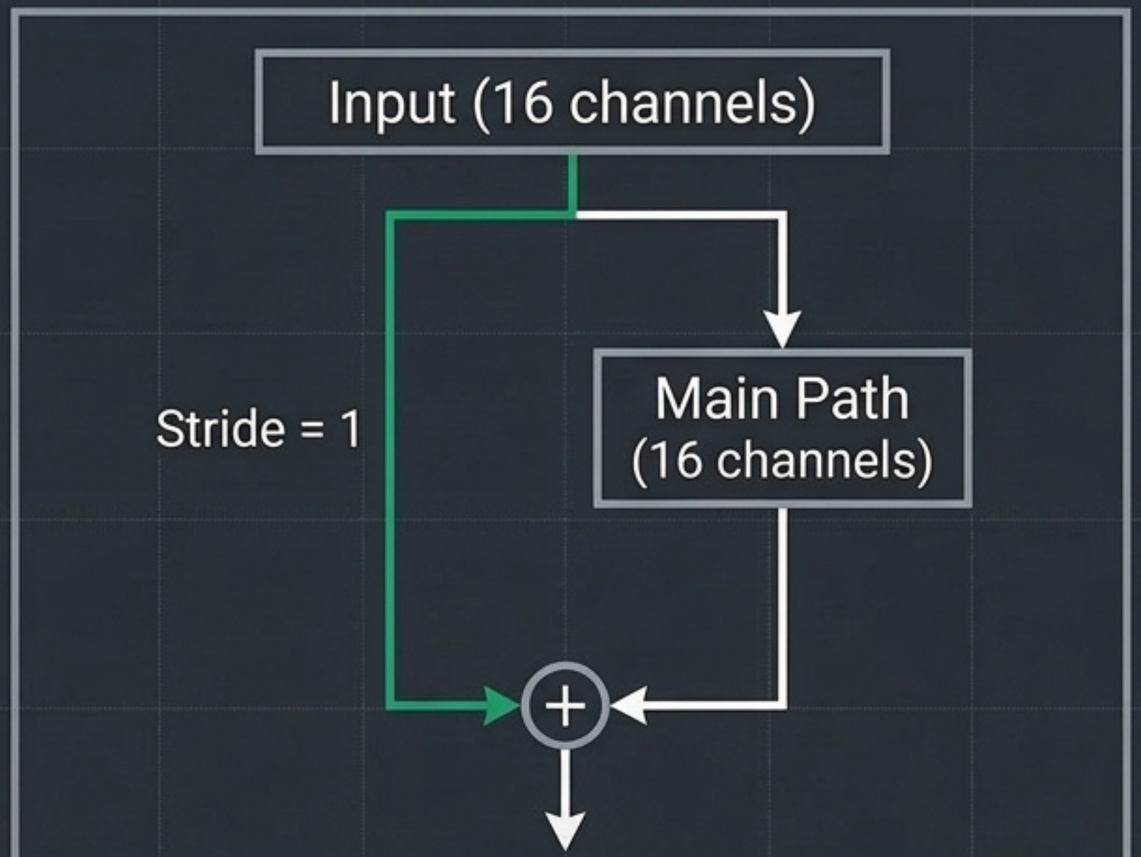
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The Residual BasicBlock: The Bypass Lane

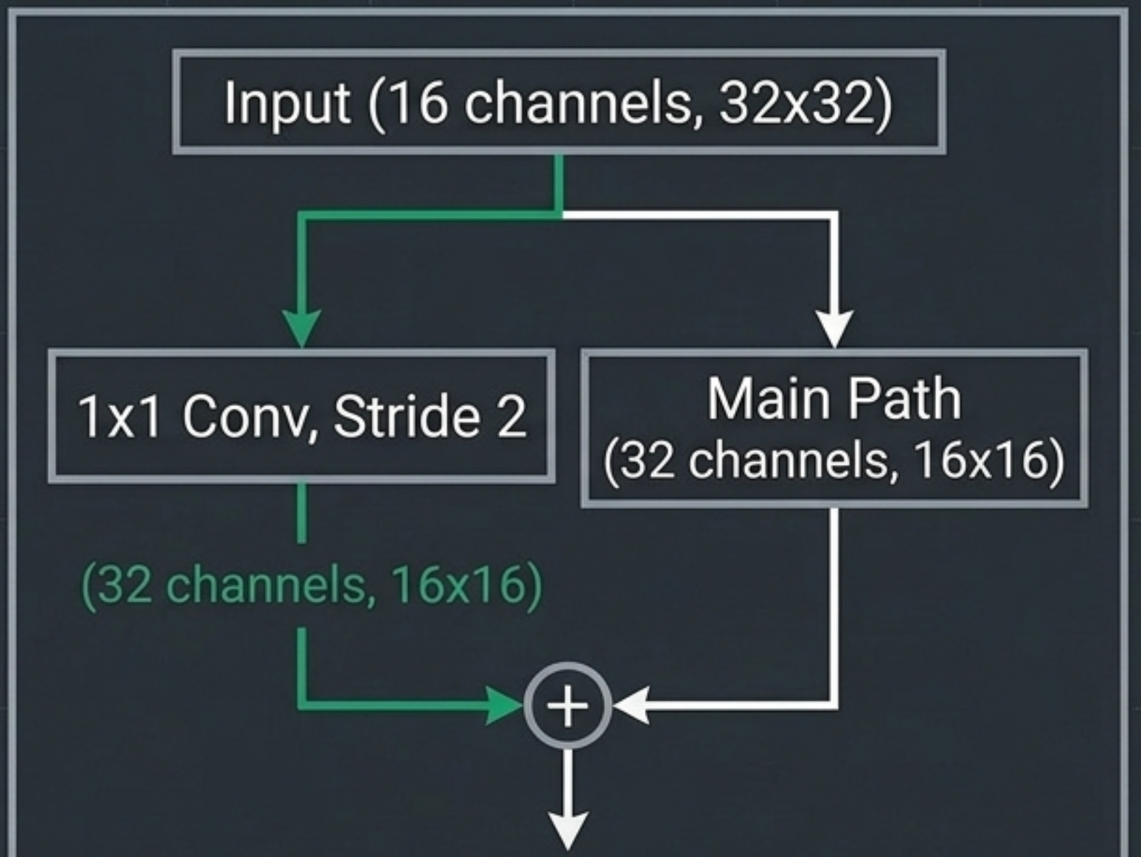
- The network learns the residual transformation ($F(x)$) rather than the raw mapping.
- The identity shortcut (x) provides a direct, uninterrupted route for the original signal.
- Gradients can flow freely backward through the skip connection, bypassing deep convolutional bottlenecks.

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Perfect Fit (Identity)

Used when channels and spatial dimensions perfectly match.

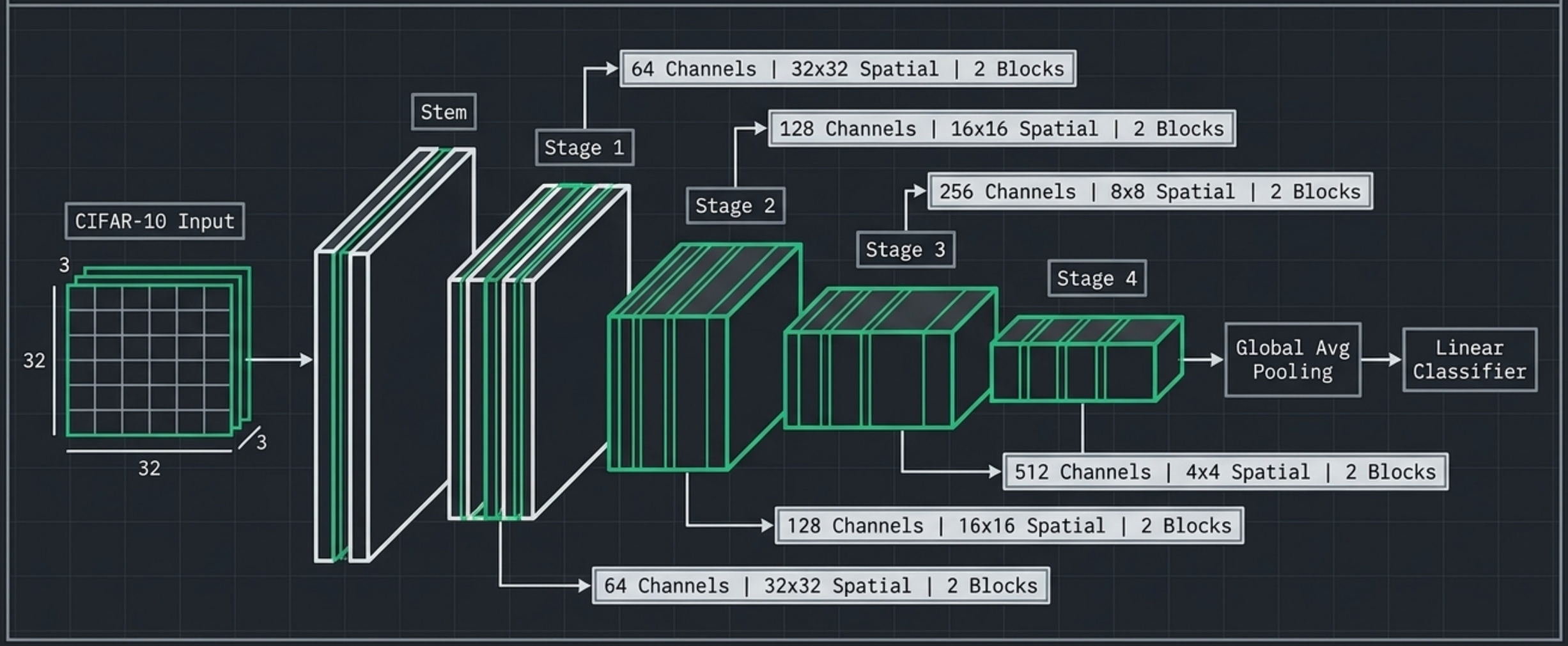


Managing Mismatch (Projection)

A 1x1 Convolution mathematically resizes the tensor to force a perfect fit for addition.

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ResNet-18 Macro-Architecture: Architectural Flow



Macro-Architecture: ResNet-18's Engine

The Progression: Each new stage strictly doubles the channel count while halving the spatial dimensions, compressing a 3-channel image into 512 dense features.

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The Architecture Matrix

Shallow PlainNet

Blocks per Stage:
[1, 1, 1, 1]

Depth:
10 Layers

Parameters:
4.73 Million

Shortcuts:
None

Deep PlainNet

Blocks per Stage:
[2, 2, 2, 2]

Depth:
18 Layers

Parameters:
11.00 Million

Shortcuts:
None

Deep ResNet-18

Blocks per Stage:
[2, 2, 2, 2]

Depth:
18 Layers

Parameters:
11.17 Million
(Due to projection weights)

Shortcuts:
Yes (Identity & Projection)

Experimental Design: The 3 Models

The Control: The 18-layer architectures are structurally identical, differing *only* in the presence of residual skip connections.

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The Laboratory Setup: Experimental Controls

Lab Specs

Dataset & Splits

CIFAR-10

45k Train / 5k Validation / 10k Hold-out Test

Optimizer Mechanics

SGD with 0.9 Momentum

Learning Rate: 0.1

Weight Decay: 5e-4

Training Budget

Batch Size: 128

Duration: 30 Epochs

Strict Controls

Identical Random Seeds (SEED = 42)

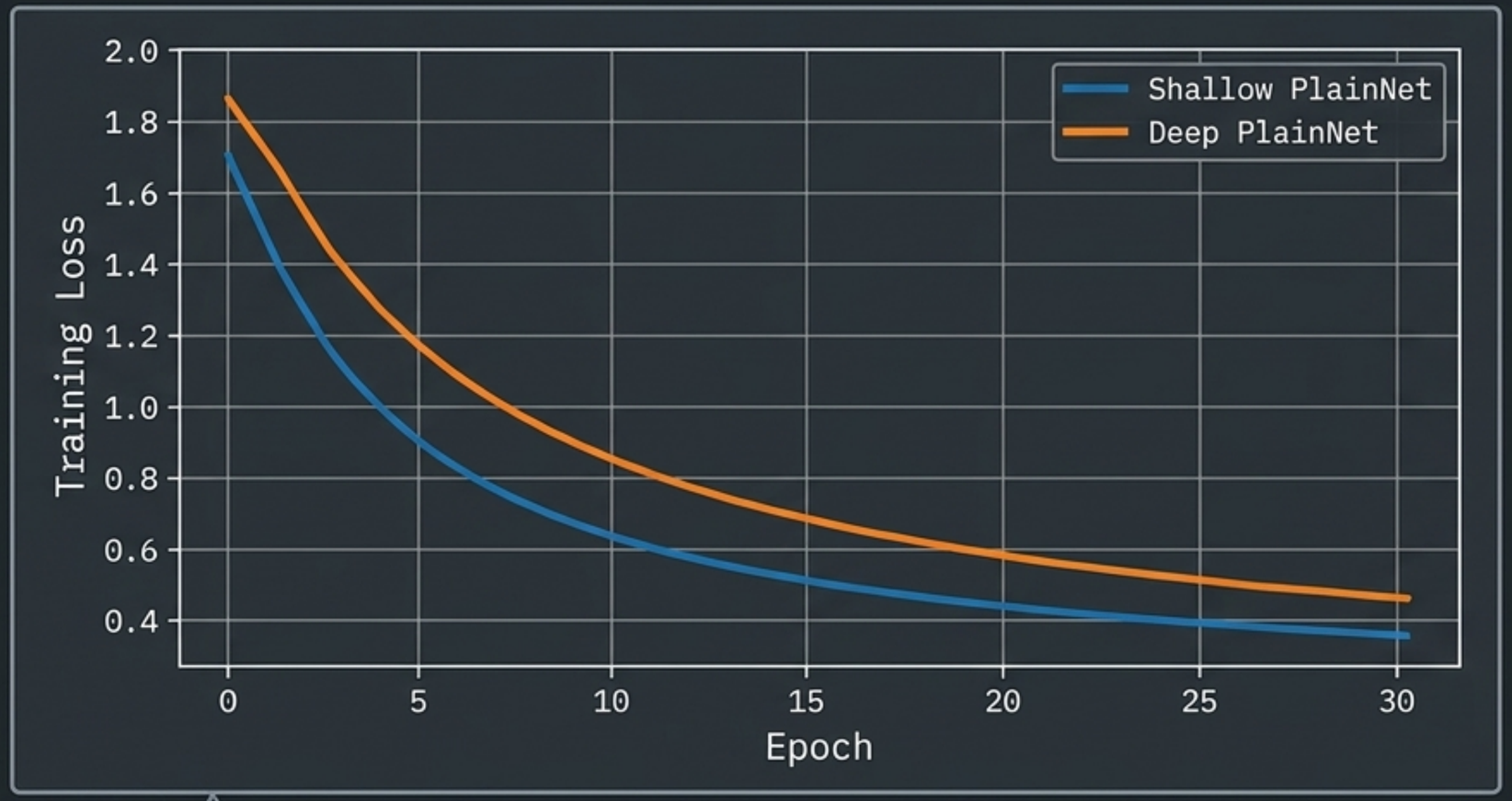
Identical Preprocessing & Augmentations

The Laboratory Setup

By locking all external variables, any divergence in learning curves is strictly attributable to architectural depth and residual routing.

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Training Loss Comparison

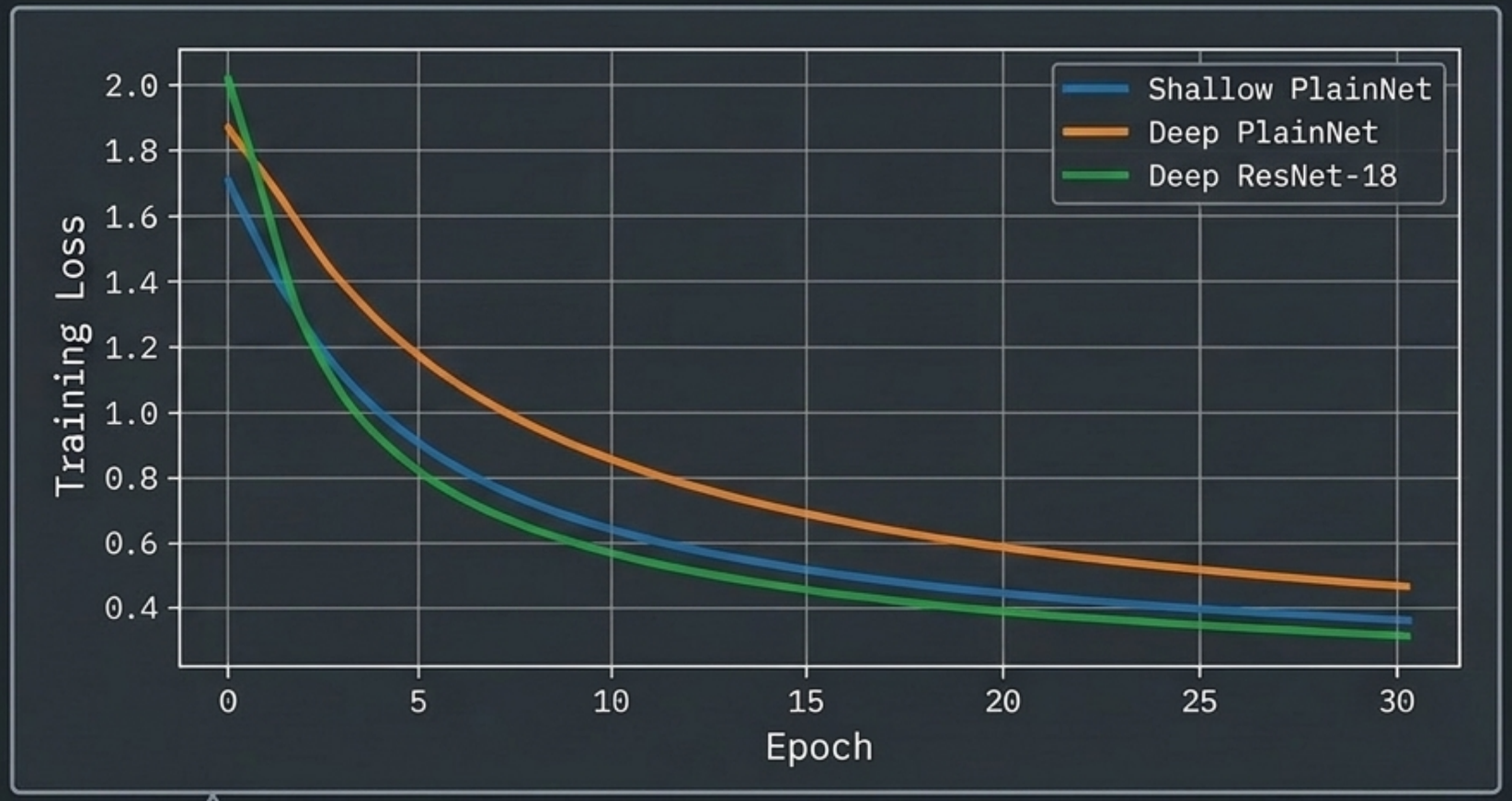


Results Phase 1: Proof of Degradation

Despite having 6.27 million more parameters and twice the depth, the 18-layer Deep PlainNet fundamentally fails to optimize as effectively as the 10-layer Shallow PlainNet. Depth alone has created an optimization bottleneck.

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Training Loss Comparison

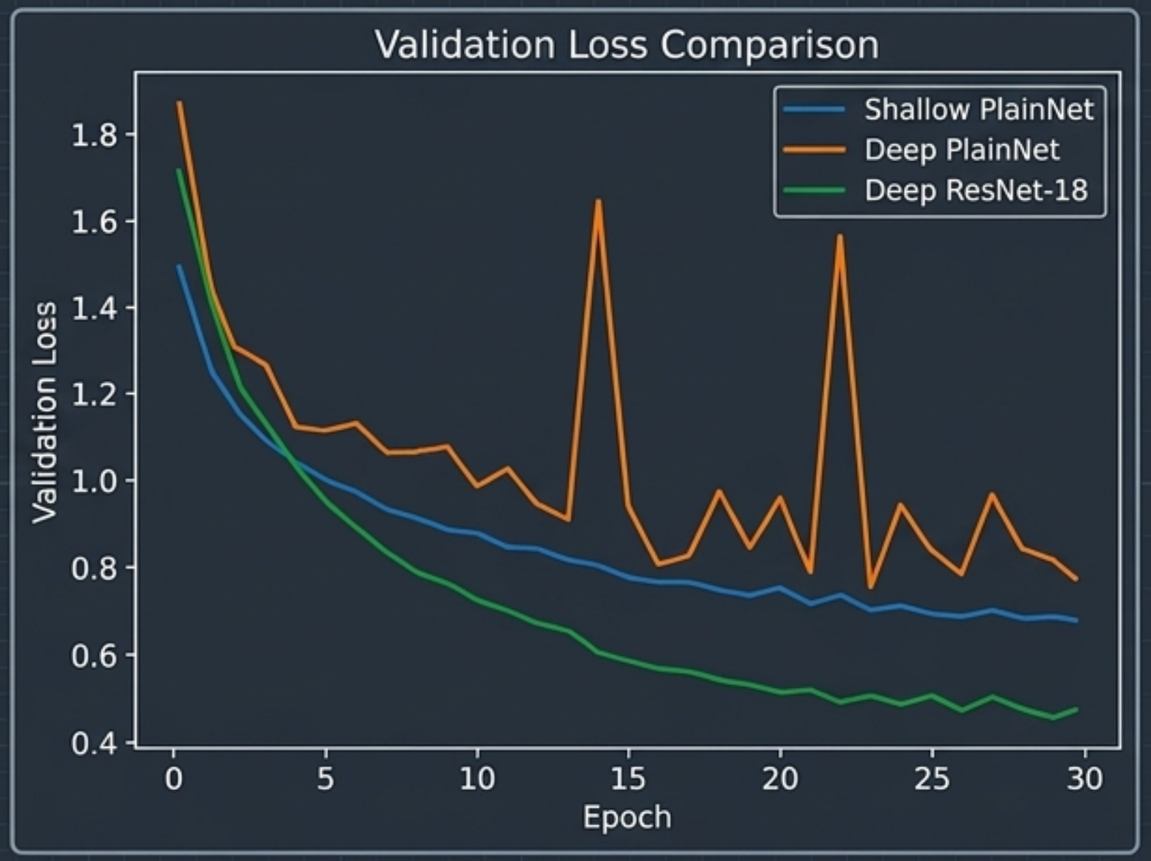


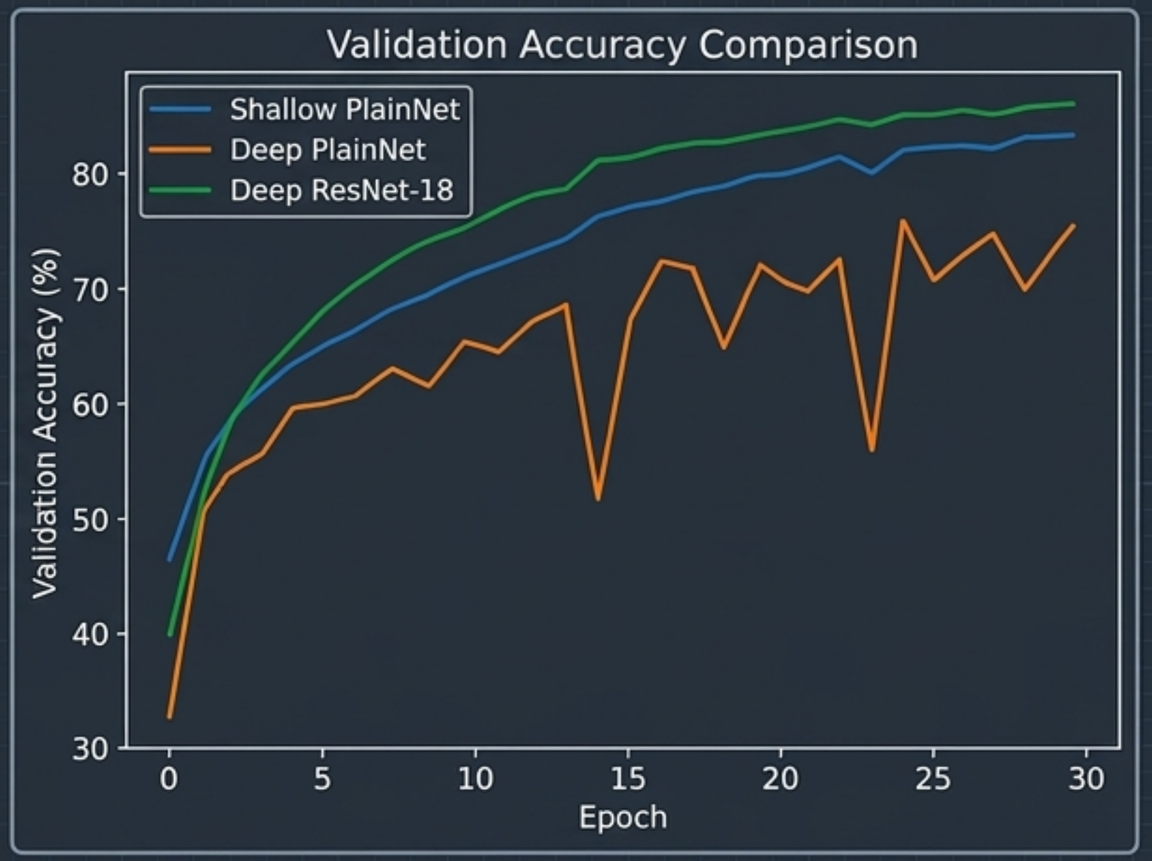
Results Phase 2: The Residual Fix

By simply adding the identity shortcut, the exact same 18-layer architecture overcomes the bottleneck. Gradients flow deeply. The Deep ResNet effectively leverages its massive parameter count to master the training data.

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Validation Performance: Generalization & Stability Analysis





Generalization & Stability

- The Deep PlainNet suffers from extreme optimization instability, failing to reliably generalize.
- The skip connections in ResNet-18 not only reduce training error but drastically smooth the optimization landscape, allowing for stable, consistent generalization.

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The Final Results Matrix

Shallow PlainNet → 83.28% Test Accuracy

(The Degradation Drop)

Deep PlainNet → 80.27% Test Accuracy

(The Residual Triumph)

Deep ResNet-18 → 84.82% Test Accuracy

Synthesis: The Final Verdict

The 4.55% accuracy gap between the two 18-layer networks is exclusively attributable to the residual connections. ResNet-18 effectively escapes the degradation trap to beat both the deep and shallow plain baselines.

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1 The Trap

Depth does not automatically equal performance. Without architectural interventions, adding layers creates severe optimization bottlenecks (Degradation).



2 The Key

Residual connections (BasicBlocks) act as bypass lanes, allowing original signals and gradients to safely pass through deep convolutional stacks.



3 The Proof

Empirical testing on CIFAR-10 definitively proves that ResNet-18 leverages its skip connections to optimize deeply, vastly outperforming plain networks of equivalent depth.



Optimization requires paths of least resistance.

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